

WEEK_6:-TASK2

MILKA NAVEENA

TASK 2 — DATA COLLECTION & DATA PREPROCESSING

2.1 Dataset Selection

For this project, I selected the **Student Performance Dataset** from the UCI Machine Learning Repository.

The dataset contains student information such as:

- School
- Gender
- Age
- Study time
- Previous failures
- Absences
- Mother's education
- Father's education
- Social activities
- Health
- Grades

The final grade is represented by **G3**.

I will use G3 to create my classification target.

2.2 Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier

from sklearn.metrics import (
    accuracy_score,
    precision_score,
    recall_score,
    confusion_matrix,
    classification_report
)

import warnings
```

Output

There should be **no error message**.

2.3 Load Dataset

```
df = pd.read_csv("student-mat.csv", sep=";")

print("Dataset loaded successfully!")
```

Output

```
Dataset loaded successfully!
```

2.4 Display First Five Records

```
df.head()
```

Output

school	gender	age	address	famsize	studytime	failures	absences	G1	G2	G3
GP	F	18	U	GT3	2	0	6	5	6	
GP	F	17	U	GT3	2	0	4	5	5	
GP	F	15	U	LE3	2	3	10	7	8	
GP	F	15	U	GT3	3	0	2	15	14	
GP	F	16	U	GT3	2	0	4	6	10	

2.5 Display Last Five Records

```
df.tail()
```

2.6 Check Number of Rows and Columns

```
print("Number of rows:", df.shape[0])  
print("Number of columns:", df.shape[1])
```

Output

```
Number of rows: 395  
Number of columns: 33
```

2.7 Check Data Types

```
df.dtypes
```

Output

school	object
gender	object
age	int64

address	object
studytime	int64
failures	int64
absences	int64
G1	int64
G2	int64
G3	int64

Explanation

The dataset contains both:

- **Categorical variables** → object
- **Numerical variables** → int64

2.8 Check Missing Values

```
missing_values = df.isnull().sum()

print(missing_values)
```

```
print("Total missing values:", df.isnull().sum().sum())
```

output

```
Total missing values: 0
```

Report Explanation

I checked the dataset for missing values using the `isnull()` function. The dataset did not contain missing values, so no imputation was required.

2.9 Check Duplicate Records

```
print("Number of duplicate records:", df.duplicated().sum())
```

```
df = df.drop_duplicates()
```

```
print("Duplicates after cleaning:", df.duplicated().sum())
```

Output

```
Duplicates after cleaning: 0
```

2.10 Statistical Summary

```
df.describe()
```

Expected Output

A statistical table containing:

- Count
- Mean
- Standard deviation
- Minimum
- 25%
- 50%
- 75%
- Maximum

for numerical columns.

2.11 Create Target Variable

The original `G3` column represents the final grade.

I will convert it into two categories:

```
G3 >= 10 → Pass  
G3 < 10  → Fail
```

Code:

```
df["performance"] = np.where(  
    df["G3"] >= 10,  
    "Pass",  
    "Fail"  
)
```

Check:

```
df[["G3", "performance"]].head(10)
```

Output

G3	performance
6	Fail
6	Fail
10	Pass
15	Pass
10	Pass
15	Pass
11	Pass
6	Fail
19	Pass
15	Pass

2.12 Select Relevant Features

```
features = [  
    "school",  
    "gender",  
    "age",  
    "studytime",  
    "failures",  
    "absences",  
    "Medu",  
    "Fedu",  
    "goout",  
    "health"  
]  
  
X = df[features]  
y = df["performance"]
```

Why these features?

These variables represent different aspects of a student's:

- Demographic information
- Study habits
- Academic history
- Attendance

- Family education
 - Social behavior
 - Health
-

2.13 Encode Categorical Variables

```
X = pd.get_dummies(X, drop_first=True)
```

```
X.head()
```

2.14 Encode Target Variable

```
label_encoder = LabelEncoder()
```

```
y = label_encoder.fit_transform(y)
```

```
print(label_encoder.classes_)
```

Output

```
['Fail' 'Pass']
```

```
Fail = 0
```

```
Pass = 1
```

2.15 Final Preprocessed Data

```
print("Feature shape:", X.shape)
print("Target shape:", y.shape)

print("\nFirst 5 feature records:")
display(X.head())

print("\nFirst 5 target values:")
print(y[:5])
```

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Rows: 395
Columns: 33
Rows: 395
Columns: 34

	school	gender	age	address	famsize	Pstatus	Medu	Fedu	Mjob	Fjob
0	GP	F	18	U	GT3	A	4	4	at_home	teacher
1	GP	F	17	U	GT3	T	1	1	at_home	other
2	GP	F	15	U	LE3	T	1	1	at_home	other
3	GP	F	15	U	GT3	T	4	2	health	services
4	GP	F	16	U	GT3	T	3	3	other	other

	...	studytime	failures	absences	G1	G2	G3
0	...	2	0	6	5	6	6
1	...	2	0	4	5	5	6
2	...	2	3	10	7	8	10
3	...	3	0	2	15	14	15
4	...	2	0	4	6	10	10

Total Missing Values: 0
Duplicate Records: 0
Duplicates After Removal: 0

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JupyterLab Python 3 (ipykernel)

```
"school",
"sex",
"age",
"studytime",
"failures",
"absences",
"Medu",
"Fedu",
"goout",
"health"
]
X = df[features]
y = df["performance"]
print("\nSelected Features:")
display(X.head())
X = pd.get_dummies(X, drop_first=True)
from sklearn.preprocessing import LabelEncoder
label_encoder = LabelEncoder()
y = label_encoder.fit_transform(y)
print("\nEncoded Features:")
display(X.head())
print("\nEncoded Target:")
print(y[:10])
print("\nFinal Feature Shape:", X.shape)
print("Final Target Shape:", y.shape)
print("\nMissing Values in Features:", X.isnull().sum().sum())
print("Missing Values in Target:", pd.isnull(y).sum())
```

Snipping Tool

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JupyterLab Python 3 (ipykernel)

```
import pandas as pd
import numpy as np
df = pd.read_csv("student-mat.csv", sep=";")
print("First 5 Records:")
display(df.head())
print("Last 5 Records:")
display(df.tail())
print("Rows:", df.shape[0])
print("Columns:", df.shape[1])
print("\nColumn Names:")
print(df.columns.tolist())
print("\nData Types:")
print(df.dtypes)
print("\nDataset Information:")
df.info()
print("\nStatistical Summary:")
display(df.describe())
print("\nMissing Values:")
print(df.isnull().sum())
print("\nTotal Missing Values:", df.isnull().sum().sum())
print("\nDuplicate Records:", df.duplicated().sum())
df = df.drop_duplicates()
print("Duplicates After Removal:", df.duplicated().sum())
df["performance"] = np.where(df["G3"] >= 10, "Pass", "Fail")
print("\nG3 and Performance:")
display(df[["G3", "performance"]].head(10))
print("\nPerformance Distribution:")
```

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